

# KHWAJA YUNUS ALI UNIVERSITY

School of Science and Engineering  
Department of Computer Science and Engineering  
Summer – 2022 Final Examination  
Program: B.Sc. in CSE (Batch – 14<sup>th</sup>) 1<sup>st</sup> Year 1<sup>st</sup> Semester  
Course Title: Physics - I  
Course Code: PHY 1101

## PART-A: MCQ

Time: 10 Minutes

Total Marks: 10x01=10

Student's ID: 

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 Invigilator's Signature: 

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(These are single best type of questions. Each of the following questions or statements has five suggested answers or completions. Put tick mark (✓) on the best answer. There is no counter marking.)

### Cognitive Knowledge

1. Which is the unit of mutual inductance?

- A) Volt
- B) H
- C)  $\text{Am}^{-2}$
- D)  $\text{JC}^{-1}$
- E)  $\text{CV}^{-1}$

### Cognitive understanding

2. The value of proportionality constant of Newman's law is .....

- A) 1
- B)  $10^{-7}$
- C)  $10^{-10}$
- D)  $\pi$
- E)  $4\pi$

### Cognitive Applying

3. What is the frequency (in Hz) of this alternating current:  $i = 200 \sin 100\pi t$  ?

- A) 50
- B) 100
- C) 120
- D) 200
- E) 220

### Cognitive Analysis

4. Which one is right for the power of the primary ( $P_p$ ) and secondary ( $P_s$ ) coil of transformer?

- A)  $P_p > P_s$
- B)  $P_p < P_s$
- C)  $P_p = P_s$
- D)  $P_p \gg P_s$
- E)  $P_p \ll P_s$

### Cognitive Applying

5. How many emf generated when a coil of 100 turns is subjected to a flux rate of 0.3 wb/sec?

- A) 0.003
- B) 0.03
- C) 0.3
- D) 3
- E) 30

**Cognitive Knowledge**

6. In case of SHO, Angular frequency,  $\omega = ?$

- A)  $K/m$
- B)  $m/K$
- C)  $\sqrt{m/K}$
- D)  $\sqrt{K/m}$
- E)  $K^2/m^2$

**Cognitive Knowledge**

7. The maximum displacement of a body, from its mean position is called .....

- A) acceleration
- B) time period
- C) frequency
- D) wave length
- E) amplitude

**Cognitive Comprehension**

8.  $x = A \sin(10\pi t + \delta)$  is the solution of the differential equation of SHM: .....

- A)  $\frac{d^2x}{dt^2} + 10\pi^2x = 0$
- B)  $\frac{d^2x}{dt^2} + 10\pi x = 0$
- C)  $\frac{d^2x}{dt^2} + 100\pi x = 0$
- D)  $\frac{d^2x}{dt^2} + 100\pi^2x = 0$
- E)  $\frac{d^2x}{dt^2} + 100x = 0$

**Cognitive Understanding**

9. Which of the following devices is used in mobile phone charger?

- A) Ammeter
- B) Photo diode
- C) Heat sensor
- D) Step-down transformer
- E) Step-up transformer

**Cognitive Understanding**

10. What amount of angle should be maintained between the current flow and the magnetic field to produce the hall effect in the conductor?

- A)  $0^\circ$
- B)  $30^\circ$
- C)  $90^\circ$
- D)  $180^\circ$
- E)  $360^\circ$

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Examiner Signature

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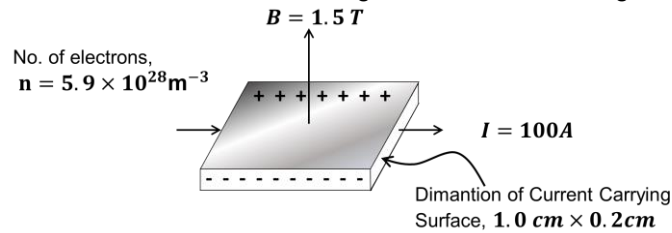
## PART-B: SAQ

Time: 1 Hour and 50 Minutes

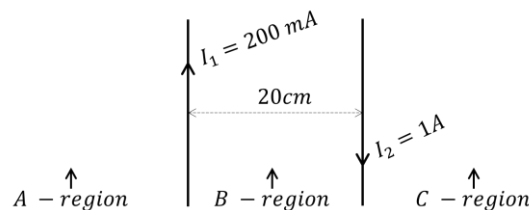
Total Marks: 5x6=30

INSTRUCTIONS: Answer question no. 01 (one) and any 04 (four) from the rest. Marks are shown by side.

1. A semiconductor ribbon carries a current from left to right, and it lies in a magnetic field as shown in figure:



- Define Hall effect. [1]
  - Identify the type of the above-mentioned semiconductor with logical explanation. [2]
  - Find the hall potential between the edges of the ribbon. [3]
2. Two parallel conductors and carrying current anti-parallel direction:



- Mention the type the force between conductors. [1]
  - What is the magnitude of the force per unit length between conductors? [2]
  - Investigate, in which region the neutral point for the magnetic field will be? Give the mathematical logic. [3]
3. a) State the Lenz's law. [1]
- b) How does electromagnetic induction obey the law of conservation of energy? [2]
- c) Magnetic flux in a coil of 400 turns changes from  $8 \times 10^{-5} wb$  to  $50 \times 10^{-5} wb$  in  $3 \times 10^{-3} sec$ . Calculate the induced  $emf$  in the coil. [3]
4. a) Why is the laminated steel core used in the transformer? [1]
- b) The radius of the circular coil is  $0.02m$  and its number of turns is 100. Calculate the magnetic field at the center of the coil if  $5A$  current flows in the coil. [2]
- c) Formulate an equation the magnetic field for a long straight current-carrying wire. [3]
5. a) Define force constant. [1]
- b) If a particle of mass  $0.05kg$  executing simple harmonic motion with frequency  $50Hz$  and if its amplitude is  $0.004m$ . Calculate its average kinetic energy. [2]
- c) Create the conditions to get maximum velocity and acceleration of simple harmonic oscillator. [3]
6. a) Write a short note on – "Lissajous Figures". [2]
- b) Analyze the following quantities from the displacement of a progressive wave at any instant of time is:  
$$x = 0.1 \sin(680\pi t - 0.30\pi)$$
  
Calculate (i) the amplitude (ii) angular frequency and (iv) time period. [4]